

Herald



Herald — §11.4.98 Full-Automation Anti-Bluff Audit

Field	Value
Audit date	2026-05-28
Authority	HelixConstitution §11.4.98 (commit 6828ff2) + Herald §108.m (commit bbf03c8)
Scope	All Herald test surfaces (integration tests, shell scripts, mutation gates, e2e invariants)
Release-gate posture	NON-COMPLIANT items have 30 days from this audit before graduating to §11.4.90 Obsolete
Deadline	2026-06-27 (T+30 days)
Methodology	Static analysis only — no test executed, no source modified
Total tests audited	794 (783 Go test/bench/example funcs + 11 shell test/e2e scripts)
COMPLIANT (hermetic)	750 Go funcs + 8 shell scripts = 758
COMPLIANT-with-creds-bootstrap	32 Go funcs + 1 shell script = 33
NON-COMPLIANT-manual-dep	2 Go funcs + 1 shell script + 1 shell script-mode = 4
STRUCTURALLY-BROKEN	0 (the former item — <code>TestDispatch_LiveClaudeInvocation</code> — was reclassified COMPLIANT-with-creds-bootstrap on 2026-05-31 after its dedicated-test-only-session-UUID fix landed; see the resolution note below)
OBSOLETE-CANDIDATE	0 (no removed code-path tests detected — all NON-COMPLIANT items are rewrite-targets, not deletion-targets)
PLAN-ONLY (out of scope)	0 (no docs/challenges/ dir exists; docs/research/ contains 4 prose-only docs counted out of scope)

Summary

Herald's test surface is overwhelmingly COMPLIANT under §11.4.98: 758 of the 794 audited tests are fully self-driving — they run hermetically with no external dependency or no human action of any kind. A further 33 tests are COMPLIANT-with-creds-bootstrap per the §11.4.98(B) explicit exception: they require one-time credential setup (Telegram bot token + chat

id in `.env`, `claude` on `PATH`, Postgres/Redis containers via Podman/Docker) — but once configured, every subsequent run is fully autonomous. This is the §11.4.98(B) permitted carve-out, NOT a manual-dep.

The audit identifies **exactly 4 NON-COMPLIANT items** that require human action during test execution itself — every one of them is a Telegram-inbound-driven test that polls `getUpdates` for a real operator-typed message. These 4 cannot run in unattended CI and are the entire §11.4.98 release-gate debt. They are:

1. `TestSubscribe_LiveBotAPI` (`commons_messaging/channels/tgram/subscribe_integration_test.go`) — 60s window for operator hand-send.
2. `TestVerticalSlice_TelegramClaudeRoundTrip` (`commons_messaging/vertical_slice_integration_test.go`) — 150s window for operator hand-send, drives the full Telegram → Claude → Telegram round-trip.
3. `tests/test_wave6_live_loop.sh` — 60s window for operator hand-send into the configured chat.
4. `tests/test_wave6.5_lifecycle.sh --manual (legacy mode)` — 15 narrated scenarios with `read -r` prompts between each one. Note: the DEFAULT mode of the same script delegates to `qaherald lifecycle` (a second Telegram bot drives the scenarios) and IS fully self-driving; only the explicit `--manual` legacy path is NON-COMPLIANT.

Update 2026-05-31 — the former STRUCTURALLY-BROKEN item is RESOLVED.

`TestDispatch_LiveClaudeInvocation` (`commons_messaging/dispatch/claude_code/dispatch_integration_test.go:39-139`) no longer accepts the conductor-colliding `HERALD_CLAUDE_SESSION_UUID`. The 2026-05-28 audit predicted the fix (dedicated test-only session UUID); the code now implements it: the test hard-rejects `HERALD_CLAUDE_SESSION_UUID` with `t.Fatalf` (lines 55-56), requires a DEDICATED `HERALD_CLAUDE_TEST_SESSION_UUID` (SKIP-with-reason when absent, lines 58-60), and hard-fails if that test UUID collides with the conductor's live anchored session (the §11.4.98 rule-2 collision guard via `d.ResolveSession()`, lines 83-84). It is therefore reclassified **COMPLIANT-with-creds-bootstrap** — once a throwaway test-only session UUID is supplied (one-time bootstrap, not human action during execution), the test is fully self-driving and cannot collide with the dev conductor's session.

The next-step plan: **Wave 8 Track B — MTProto user-client harness**. Telegram bots cannot see other bots' messages (proven blocker, `docs/research/telegram-bot-to-bot-constraint.md`), so a 2nd-bot automation driver is impossible. The remaining path is a `gotd/td` user-client that posts messages on behalf of a real Telegram user account; this drives all 4 NON-COMPLIANT items in one fix. Operator-pending: see `docs/requirements/blockers/missing_env_variables.md` for the credentials needed (`TG_API_ID`, `TG_API_HASH`, user phone, session file).

COMPLIANT tests (hermetic — no creds, no human action)

Go tests — fully hermetic (sampled by package)

These are the default-hermetic majority. Every test below runs with `go test ./...` (no `-tags=integration`), no env vars, no human input, no external services. The audit confirms this by absence: each file uses only Go stdlib, `httptest`, in-process loopback, mocks, or the `commons/stresschaos` scaffold. Total: ~750 funcs across the 129 non-integration `_test.go` files.

Test surface	Module / file pattern	Why COMPLET
Commons L0	commons/*_test.go, commons/cli/*_test.go, commons/gitops/*_test.go, commons/stresschaos/*_test.go	Pure-Go unit tests. commons/... serve_dual_h3_test.go, httptest, process Gin server network.
Prefix	commons_prefix/prefix_test.go	Algorithm tests.
Auth	commons_auth/verifier_test.go	JWT verifier tests against in-test keys.
TLS	commons_tls/cert_test.go	Self-signed certificates, parse round-trip.
Storage	commons_storage/migrations_test.go, storage_test.go, resource_stress_chaos_test.go	Migration path tests; checks; no DB.
Infra (default)	commons_infra/boot_test.go, clients_test.go, task_repository_test.go	Mock-Pool tests, _integration, siblings are tests default.
Constitution	commons_constitution/audit_test.go, bundle_test.go, cloudevents_test.go, emit_test.go, evaluator_test.go, eventbus_test.go, integration_test.go, ladder/memory_test.go, state/memory_test.go, plus audit_stress_chaos_test.go	In-memory simulation, implementation commons/... scaffold.
Messaging — channels (default)	commons_messaging/channels/channel_test.go, inbox_test.go, registry_test.go, selffilter_test.go, null/null_test.go, slack/*_test.go, tgram/attachments_test.go, tgram/export_test.go, tgram/send_ratelimit_test.go, tgram/send_reply_test.go, tgram/send_security_test.go, tgram/subscribe_test.go, tgram/subscribe_lifecycle_test.go, tgram/webhook_test.go, tgram/stress_chaos_test.go	Hermetic: in-memory, fault-injection, chaos suite; no DB.
Messaging — dispatch (default)	commons_messaging/dispatch/claude_code/bootstrap_test.go, claude_code_test.go, dispatch_stress_chaos_test.go, export_test.go	Mock Claude tests via fake binaries, t.TempDir.
pherald	all pherald/cmd/pherald/*_test.go, pherald/internal/{bindings, http, inbound, runner, wizard}/**/*_test.go (43 funcs)	Cobra unit tests, memory dispatch, fake redis/PC, fakes_test.go.
qaherald	all qaherald/cmd/qaherald/*_test.go, qaherald/internal/{herald, lifecycle, messenger, report, scenario, transcript}/**/*_test.go	Unit tests + benchmarks; the harness; the integration gated behind checks.
Flavor binaries	{bherald, cherald, iherald, rherald, scherald, sherald}/cmd/**/*_test.go, {...}/internal/**/*_test.go	Cobra unit tests, compliance/benchmarks, all hermetic.

Shell scripts — fully hermetic (no creds, no human action)

Script

tests/
test_constitution_inheritance.sh

tests/
test_constitution_inheritance_meta.sh

tests/test_i6_refinement_meta.sh

tests/test_i8_usability_meta.sh

tests/test_wave2_mutation_meta.sh

tests/test_wave3_mutation_meta.sh

tests/test_wave4_mutation_meta.sh

tests/test_wave4b_mutation_meta.sh

tests/test_wave6_mutation_meta.sh

tests/test_wave6.5_mutation_meta.sh

tests/test_wave7_mutation_meta.sh

tests/
test_stress_chaos_mutation_meta.sh

Why COMPLIANT

Parent-walk + grep against checked-in files.

Paired §1.1 mutation gate — snapshots → mutates Constitution.md → asserts gate FAILs → restores. Autonomous.

Paired §1.1 mutation gate against .gitmodules. Autonomous.

Paired §1.1 mutation gate against CLAUDE.md / AGENTS.md / HERALD_CONSTITUTION.md. Autonomous.

Paired §1.1 mutation gate against flavor binaries (sherald/cherald/iherald/bherald/rherald/scherald) + e2e_bluff_hunt invariants E19-E33. Autonomous.

Paired §1.1 mutation gate for Wave 3a JWT verifier + pherald Runner pipeline (E35-E47). Autonomous.

Paired §1.1 mutation gate for Wave 4a HTTP/3 + Brotli + Alt-Svc + TLS-1.3 (E49-E55). Autonomous.

Paired §1.1 mutation gate for Wave 4b TOON content negotiation (E56-E62). Autonomous.

Paired §1.1 mutation gate for Wave 6 inbound runtime + CC headless bridge. Hermetic detectors only (per file comment: "no live Telegram / claude binary needed"). Autonomous.

Paired §1.1 mutation gate for Wave 6.5 ticket-lifecycle (T1..T6). Hermetic. Autonomous.

Paired §1.1 mutation gate for Wave 7 generic messenger framework (registry + selffilter + Slack adapter). Hermetic. Autonomous.

Paired §1.1 mutation gate for the GAP-3 §11.4.85 stress + chaos suite (HRD-130, 5 mutations). Hermetic detectors. Autonomous.

COMPLIANT-with-creds-bootstrap tests (§11.4.98(B) explicit exception)

These require one-time credential bootstrap OUTSIDE test execution. Once configured (`.env` + `~/ .bashrc` exports + `claude` on `PATH` + container runtime up), every subsequent run is fully autonomous. This is the §11.4.98(B) permitted exception and is **COMPLIANT** per the rule.

Go integration tests (creds-bootstrap, fully autonomous once configured)

Test	File
TestHealthCheck_LiveBotAPI	<code>commons_messaging/channels/tg/healthcheck_integration_test.go</code>
TestSend_LiveBotAPI	<code>commons_messaging/channels/tg/send_integration_test.go</code>
TestSend_PersistsDeliveryEvidence	<code>commons_messaging/channels/tg/persist_integration_test.go</code>
TestBootstrapSession_LiveClaudeInvocation	<code>commons_messaging/dispatch/claude_code/bootstrap_integration_test.go</code>
TestDispatch_LiveClaudeInvocation	<code>commons_messaging/dispatch/claude_code/dispatch_integration_test.go</code>
TestDispatch_PersistsSessionState	<code>commons_messaging/dispatch/claude_code/persist_integration_test.go</code>
TestLiveBoot_PostgresOnly	<code>commons_infra/boot_integration_test.go</code>
TestUp_PopulatesPool	<code>commons_infra/clients_integration_test.go</code>
TestUp_PopulatesRedis_TTLRoundTrip	<code>commons_infra/clients_integration_test.go</code>
TestUp_PopulatesQueue_EnqueueDequeueRoundTrip	<code>commons_infra/clients_integration_test.go</code>
TestRepo*_RoundTrip (×14)	<code>commons_infra/task_repository_integration_</code>

Test	File
TestRLS_TenantIsolation_RoundTrip	commons_storage/ storage_integration_test.go
TestPostgresStore_RecordAndGet + 5 siblings	commons_constitution/ postgres_integration_test.go
TestClient_LiveGetMe	qaherald/internal/tgram/ client_test.go
TestTgram_Chaos_GetUpdatesPollerResilience	commons_messaging/channels/t/ stress_chaos_test.go

Shell script (creds-bootstrap, fully autonomous once configured)

Script	Bootstrap requirement	Why COMPLIANT
tests/ test_resource_stress_chaos.sh	Podman/Docker + opt-in HERALD_STRESS_LIVE_OOM=1	The opt-in flag is a host-safety gate (§12.6) — not a manual-dep during execution. Once set, the script runs container OOM + connection-churn scenarios autonomously. SKIPs-with-reason if creds/flag absent (per §11.4.3). DEFAULT mode delegates the 15-scenario lifecycle to qaherald lifecycle — the qa-bot posts each
tests/ test_wave6.5_lifecycle.sh (DEFAULT mode)	HERALD_TGRAM_BOT_TOKEN + HERALD_TGRAM_CHAT_ID + HERALD_QA_BOT_TOKEN (2nd Telegram bot) + HERALD_OPERATOR_IDS + claude on PATH	scenario's input via Telegram, reads pherald-bot's replies, asserts wire-byte evidence, all without human typing. The -- manual flag flips to the legacy operator-typing

Script	Bootstrap requirement	Why COMPLIANT
		path (see NON-COMPLIANT below). The --manual flag flips to the legacy operator-typing path (see NON-COMPLIANT row below).

E-invariants in scripts/e2e_bluff_hunt.sh (88 total, E1-E88)

scripts/e2e_bluff_hunt.sh is the canonical end-to-end smoke. Of its 88 invariants (E1-E88):

- **E1-E12, E19-E33, E35-E48, E49-E62, E81-E88** (~70 invariants): hermetic or creds-bootstrap. COMPLIANT.
- **E13-E18** (6 invariants): creds-bootstrap (live PG / Redis / Telegram outbound / Claude Code). COMPLIANT.
- **E34** (vertical slice): manual-dep — wraps TestVerticalSlice_TelegramClaudeRoundTrip. The e2e harness SKIPs E34 unless HERALD_TGRAM_LIVE_INBOUND=1 is set (signalling an attended session). Classified NON-COMPLIANT via the underlying Go test.
- **E63-E70** (Wave 6 inbound): currently SKIP-with-reason pending T10b live closed-loop evidence; E70 explicitly gated behind HERALD_W6_LIVE_LOOP=1 because tests/test_wave6_live_loop.sh is an ATTENDED test. Classified NON-COMPLIANT via the underlying script.
- **E71-E80** (Wave 6.5 lifecycle): currently SKIP-with-reason pending the full S1..S15 evidence dir. The new automated mode (qa Herald lifecycle) will close these once an operator runs it; until then SKIP.

NON-COMPLIANT-manual-dep tests (require human action during execution)

Test	File
TestSubscribe_LiveBotAPI	commons_messaging/channels/tgram/subscribe_integration_test.go (lines 30-58)
TestVerticalSlice_TelegramClaudeRoundTrip	commons_messaging/vertical_slice_integration_test.go (lines 70-280)

Test

File

tests/test_wave6_live_loop.sh

tests/test_wave6_live_loop.sh (lines 53-58 narrate the precondition; lines 64-92 poll for the message)

tests/test_wave6.5_lifecycle.sh --manual (legacy mode only)

tests/test_wave6.5_lifecycle.sh (lines 156-346 — --manual mode; line 210 read -r _ after each narrate prompt; 15 such prompts S1..S15)

STRUCTURALLY-BROKEN (none — the one former item is RESOLVED)

The 2026-05-28 audit carved out a single STRUCTURALLY-BROKEN item, TestDispatch_LiveClaudeInvocation. It was fixed and reclassified COMPLIANT-with-creds-bootstrap on 2026-05-31 — the count above is now 0. The table below is retained as a forensic record of the original finding and its resolution.

Test

File

Original

TestDispatch_LiveClaudeInvocation

commons_messaging/dispatch/
claude_code/
dispatch_integration_test.go:39-139

FAILED
stdout v
HERAL
equal to
dev-ses
2026-05-
LIVE-
05_di
after 18

OBSOLETE-CANDIDATE (none detected)

No NON-COMPLIANT test exercises a removed or superseded code path. All 4 NON-COMPLIANT items (plus the former STRUCTURALLY-BROKEN item, now resolved + reclassified COMPLIANT-with-creds-bootstrap on 2026-05-31) target production code that is live and load-bearing (`tgram.Subscribe` long-poll loop, `claude_code.Dispatch`, `pherald listen` inbound runtime, the full §32.6 ticket-lifecycle pipeline). The rewrite path is **swap the driver, keep the assertions** — not delete-and-replace.

If, after Wave 8 Track B lands, any of the legacy operator-typing prompts in `tests/test_wave6.5_lifecycle.sh --manual` is decided to be a maintenance burden, it should graduate to §11.4.90 *Obsolete* (→ *Fixed.md*) with Reason *superseded-by-design-change* and *Superseding-item* pointing at the qaherald-driven automated mode.

PLAN-ONLY (out of scope — for completeness)

`docs/research/` carries 4 prose-only research documents with NO executable tests:

- `docs/research/constitution-compliance-audit-2026-05-27.md`
- `docs/research/hrd-obsolescence-and-qa-coverage-audit-2026-05-27.md`
- `docs/research/telegram-bot-to-bot-constraint.md` (the structural blocker cited above)
- `docs/research/workable-items-phase2-assessment-2026-05-27.md`

These emit no PASS/FAIL and are correctly out of §11.4.98 scope. `docs/research/protocols/` (58 sub-docs) is also prose-only and out of scope.

No `docs/challenges/` directory exists in this repo.

Release-gate timeline

Date

Milestone

Audit published. NON-COMPLIANT items: 4. Release-gate clock starts.

Date	Milestone
2026-05-28 (today)	
2026-06-05 (T+8d)	Wave 8 Track B planning checkpoint — MTProto harness HRD-NNN filed; operator credentials request (TG_API_ID, TG_API_HASH, user phone) acknowledged.
2026-06-13 (T+16d)	Wave 8 Track B implementation midpoint — harness skeleton + 1 of 4 NON-COMPLIANT items rewritten end-to-end (canonical target: TestSubscribe_LiveBotAPI).
2026-05-31 (T+3d, landed early)	STRUCTURALLY-BROKEN TestDispatch_LiveClaudeInvocation FIXED (dedicated test-only session UUID, conductor-collision guard) and reclassified COMPLIANT-with-creds-bootstrap — ahead of the predicted T+23d closeout. STRUCTURALLY-BROKEN count → 0.
2026-06-20 (T+23d)	Wave 8 Track B implementation closeout — all 4 NON-COMPLIANT items rewritten. (The TestDispatch_LiveClaudeInvocation fix originally bundled here already landed 2026-05-31, see above.)
2026-06-27 (T+30d)	§11.4.98 release-gate deadline. All 4 NON-COMPLIANT items MUST either: (a) be rewritten autonomous and PASS in CI, OR (b) be migrated to §11.4.90 Obsolete (→ Fixed.md) with full forensic detail (Since, Reason, Superseding-item, Triple-check evidence per §11.4.6). No third option exists.

Blocker on the timeline: Wave 8 Track B is blocked on operator-supplied Telegram user-account credentials. The blocker is fully documented at docs/requirements/blockers/missing_env_variables.md. Until those credentials land, the §11.4.3 SKIP-with-reason posture holds — the audit honestly reports the items as NON-COMPLIANT-but-honestly-skipped, not as PASS-bluffs.

Cross-references

Anchor	Path / commit
HelixConstitution §11.4.98 (canonical)	../constitution/Constitution.md commit 6828ff2
Herald §108.m (project-binding restatement)	docs/guides/HERALD_CONSTITUTION.md r8 §108.m, commit bbf03c8
Live-evidence README (2026-05-28 run)	docs/qa/HRD-LIVE-20260528T082128Z/README.md
Missing-env blocker	docs/requirements/blockers/missing_env_variables.md
Telegram bot-to-bot wall (Wave 7→8 driver constraint)	docs/research/telegram-bot-to-bot-constraint.md
§107 covenant (Herald)	CLAUDE.md §"End-user-usability covenant" + docs/guides/HERALD_CONSTITUTION.md §107
§107.x docs/qa evidence mandate	CLAUDE.md §107.x + Helix §11.4.83
§107.y working-tree quiescence (mutation-gate safety)	CLAUDE.md §107.y + Helix §11.4.84
§11.4.90 Obsolete status mechanism	Helix §11.4.90, restated in CLAUDE.md
	Helix §11.4.4 (PASS-bluff discovery halts the cycle)

Anchor	Path / commit
§11.4.4 test-interrupt-on-discovery	
§11.4.3 explicit SKIP-with-reason	Helix §11.4.3 — the only honest non-PASS posture in absence of dependency

Audit posture statement. Herald is, to within the 4 NON-COMPLIANT items above, §11.4.98-compliant today (the former 1 STRUCTURALLY-BROKEN item was fixed + reclassified COMPLIANT-with-creds-bootstrap on 2026-05-31). The 4 outstanding items are fully scoped to a single Wave (8 Track B) and a single dependency (operator-supplied Telegram user-account credentials). No structural rearchitecture is required; the rewrites swap drivers and keep assertions. The release-gate clock is honest: 30 days, no extension hatch, two paths (rewrite OR §11.4.90 Obsolete). The DEFAULT path of `tests/test_wave6.5_lifecycle.sh` (automated qaherald-driven) is COMPLIANT today; only its explicit legacy `--manual` mode counts toward the debt.

— end of audit —